



Koneru Lakshmaiah Education Foundation

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Sec : 8 DBSE -DBD

MySQL Views

MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

1. What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.
- Whenever you query the view, MySQL retrieves the latest data from the underlying tables.

2. Setup: Employees and Departments Tables

Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeedb1;  
USE employeedb1;
```

```
CREATE TABLE IF NOT EXISTS departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE IF NOT EXISTS employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    emp_salary INT,  
    emp_dept_id INT  
);
```

Automatic context help is disabled. Use the toolbar to move to the current caret position or to toggle automatic context help.

```
1 • CREATE DATABASE IF NOT EXISTS employeeedb1;
2
3 • USE employeeedb1;
4 • CREATE TABLE IF NOT EXISTS departments (
5     dept_id INT PRIMARY KEY,
6     dept_name VARCHAR(50)
7 );
8 • CREATE TABLE IF NOT EXISTS employees (
9     emp_id INT PRIMARY KEY,
10    emp_name VARCHAR(50),
11    dept_id INT,
12    salary DECIMAL(10,2),
13    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
14 );
15 • INSERT INTO departments VALUES
16     (1, 'CSE'),
17     (2, 'ECE'),
18     (3, 'EEE');
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 14	13:48:58	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), ...	0 row(s) affected
⚠ 15	13:50:00	CREATE DATABASE IF NOT EXISTS employeeedb1	1 row(s) affected, 1 warning(s): 1007 Can't create database 'employeeedb1'; database exists
✓ 16	13:50:00	USE employeeedb1	0 row(s) affected
⚠ 17	13:50:00	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50))	0 row(s) affected, 1 warning(s): 1050 Table 'departments' already exists
⚠ 18	13:50:00	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), ...	0 row(s) affected, 1 warning(s): 1050 Table 'employees' already exists
✓ 19	13:50:00	INSERT INTO departments VALUES (1, 'CSE'), (2, 'ECE'), (3, 'EEE')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
19 • INSERT INTO employees VALUES
20     (101, 'Rahul', 1, 50000),
21     (102, 'Priya', 2, 65000),
22     (103, 'Anil', 1, 55000),
23     (104, 'Sneha', 3, 70000),
24     (105, 'Kiran', 2, 48000);
25 • SELECT * FROM employees;
```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
*	NULL	NULL	NULL	NULL

employees 1 x

Apply Revert

Result Grid

Form Editor

Field Types


```

34
35 • CREATE VIEW HighSalaryEmployees AS
36 SELECT emp_id, emp_name, salary
37 FROM employees
38 WHERE salary > 55000;
39
40 • SELECT * FROM HighSalaryEmployees;

```

Result Grid

Filter Rows:

Export: Wrap Cell Content:

	emp_id	emp_name	salary
▶	102	Priya	65000.00
	104	Sneha	70000.00

Result Grid

Form Editor

Field Types

employees 4 EmployeeView 5 HighSalaryEmployees 6 x Read Only Context

Output

Action Output

#	Time	Action
✓ 65	13:54:20	DROP VIEW IF EXISTS EmployeeView
✓ 66	13:54:20	CREATE VIEW EmployeeView AS SELECT * FROM employees
✓ 67	13:54:20	SELECT * FROM EmployeeView LIMIT 0, 1000
⚠ 68	13:54:20	DROP VIEW IF EXISTS HighSalaryEmployees
✓ 69	13:54:20	CREATE VIEW HighSalaryEmployees AS SELECT emp_id, emp_name, salary FROM employees WHERE sala...
✓ 70	13:54:20	SELECT * FROM HighSalaryEmployees LIMIT 0, 1000

Limit to 1000 rows

```
47      d.dept_name,  
48      e.salary  
49  FROM employees e  
50  JOIN departments d  
51  ON e.dept_id = d.dept_id;  
52  
53 • SELECT * FROM EmployeeDepartmentView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00
	102	Priya	ECE	65000.00
	105	Kiran	ECE	48000.00
	104	Sneha	EEE	70000.00

employees 7 EmployeeView 8 HighSalaryEmployees 9 EmployeeDepartmentView 10 Read Only

```
61      e.salary  
62  FROM employees e  
63  JOIN departments d  
64  ON e.dept_id = d.dept_id  
65  WHERE d.dept_name = 'CSE';  
66  
67 • SELECT * FROM CSEEmployees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	50000.00
	103	Anil	CSE	55000.00

```

74      AVG(e.salary) AS AverageSalary
75  FROM employees e
76  JOIN departments d
77  ON e.dept_id = d.dept_id
78  GROUP BY d.dept_id, d.dept_name;
79
80 • SELECT * FROM DepartmentSalaryView;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Result Grid
dept_name	EmployeeCount	AverageSalary		
CSE	2	52500.000000		
ECE	2	56500.000000		
EEE	1	70000.000000		

Form Editor
Field Types

Employees 18 EmployeeDepartmentView 19 CSEEmployees 20 Department Read Only x

```

87  FROM employees e
88  JOIN departments d
89  ON e.dept_id = d.dept_id
90  GROUP BY d.dept_id, d.dept_name
91  HAVING AVG(e.salary) > 55000;
92
93 • SELECT * FROM HighAverageDepartments;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Result Grid
dept_name	AverageSalary			
ECE	56500.000000			
EEE	70000.000000			

Form Editor
Field Types

artmentView 25 CSEEmployees 26 DepartmentSalaryView 27 HighAverageD Read Only x

```
94 • DROP VIEW IF EXISTS EmployeeCountView;
95
96 • CREATE VIEW EmployeeCountView AS
97   SELECT COUNT(*) AS TotalEmployees
98   FROM employees;
99
100 • SELECT * FROM EmployeeCountView;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	TotalEmployees
▶	5

Result
Grid

Form
Editor

Field
Types

oyees 33

DepartmentSalaryView 34

HighAverageDepartments 35

Employee Read Only

